Classical action selection: units, symmetries and orbital thresholds

Classical mechanics admits positive actions accumulating at zero even in one fixed smooth, confining, force-bounded Hamiltonian. Positive orbital thresholds survive under additional excitation or singular-core assumptions: a regular closed orbit with speed at least $v_* > 0$ and force at most $F_{\max}$ has the sharp bound $J \geq m^2 v_*^3 / F_{\max}$; the relativistic Coulomb model has $|L| > k/c$ precisely on its complete collision-free bound domain. These are conditional classical bounds. Neither identifies a universal scale with physical $\hbar$ or constructs quantum states, dynamics or instruments.

The logical order is to specify the observable and admitted class, determine its fixed dimensional constants, test its scaling symmetries, and only then ask which dynamical restriction excludes zero. This treatment incorporates C052–C061 and Q14. The written proofs and historical model/coordinator checks are internal checks; the dimensional and orbital ingredients are established mathematics, and no literature novelty is claimed.

Observables and the admissible class

Keep $h = 2\pi\hbar$ explicit. An action unit is not an action floor, and several quantities with action units have different meanings:

Quantity	Definition	Dependence that must be specified
Action excess	$S[q+\eta] - S[q]$	Endpoint and comparison-path class
Orbital action	$J = (2\pi)^{-1} \oint p \cdot dq$	Regular trajectory and canonical momentum
Covariance action	$2\sqrt{\det \mathrm{Cov}(x, p_x)}$	Preparation and selected canonical plane
Displacement response	$m \,\mathrm{Var}[x(t+\Delta) - x(t)]/\Delta$	Preparation, window and order of limits
Reconstruction product	Separate scalar minimax errors $R_x R_P$	Records, calibration, precision and delay

A spectral gap belongs to a specified operator and has energy or inverse-time units. The susceptibility treatment and field-theory comparison explain the extra identifications needed to relate it to these observables. The classical apparatus, calibration and bounded-force proofs retain the distinct information constraints behind reconstruction products.

Even action excess closes continuously without changing the dynamics used to define the action. For a constant-force extremal on $[0, T]$, a fixed-endpoint variation $\eta(t) = at(T-t)$ has

$$S[q_* + \eta] - S[q_*] = \frac{m}{2} \int_0^T \dot{\eta}^2 dt = \frac{m a^2 T^3}{6} \longrightarrow 0.$$

The comparison paths need not be solutions of the boundary-value problem. A positive Dirichlet Hessian controls cost relative to variation norm; it does not bound that norm away from zero. The independently useful foundations manuscript retains the smooth-potential and relativistic speed-margin extensions, the normalized Hessian estimate, and the free-kernel expansion. Its quantum finite-copy threshold assumes a supplied phase unit and measurement rule; the exact controlled-copy law also appears in Galileo interference.

Dimensional necessity and a fixed Coulomb unit

Specify positive fixed constants $c_1, \ldots, c_r$, admitted masses, preparations and solutions. Let $d_i \in \mathbb{R}^3$ be their mass–length–time dimension vectors, $d_A = (1, 2, -1)$ the action vector, and $\mathcal{A}(c)$ the set of attained nonnegative actions. Define

$$g(c) = \inf\{A \in \mathcal{A}(c) : A > 0\},$$

with $g = \infty$ if no positive action is attained. Assume dimensional homogeneity and that the infimum ranges over every non-fixed mass and preparation parameter. For a change of units $\lambda \in (0, \infty)^3$,

$$\mathcal{A}(\lambda^{d_1} c_1, \ldots, \lambda^{d_r} c_r) = \lambda^{d_A} \mathcal{A}(c).$$

Dimensional theorem. If d_A is outside the span of the d_i, then $g \in \{0, \infty\}$. Otherwise, for $d_A = \sum_i a_i d_i$,

$$g(c) = \left(\prod_i c_i^{a_i} \right) F(\pi_1, \ldots, \pi_s),$$

where the π_j are a basis of dimensionless products and F takes values in $[0, \infty]$. If the d_i are independent, F is a constant.

Proof. Write $\lambda = e^{\ell}$. In the first case choose ℓ orthogonal to every d_i but with $\ell \cdot d_A \neq 0$. Constants remain fixed while $g = e^{t\ell \cdot d_A} g$ for all real t, forcing zero or infinity. In the second case divide g by the displayed action product; the quotient is invariant under unit changes and hence depends only on the dimensionless products. This is the Buckingham reduction. □

For the specified constant set $k_e = e^2/(4\pi\epsilon_0)$, c and G, the dimension vectors are $(1, 3, -2)$, $(0, 1, -1)$ and $(-1, 3, -2)$. Solving

$$a(1, 3, -2) + b(0, 1, -1) + \gamma(-1, 3, -2) = (1, 2, -1)$$

gives uniquely $a = 1$, $b = -1$, $\gamma = 0$. Thus k_e/c is the unique monomial action unit for this constant set; G, c alone supply none. Including a fixed mass adds Gm^2/c. A fixed length, mass and c supply $m_0 c \ell_0$. These statements depend on the declared constants, not on an exhaustive list of what classical physics could contain.

For a chosen elementary charge, $k_e/c = \alpha\hbar$, with $\alpha = e^2/(4\pi\epsilon_0\hbar c)$. This is a comparison of units with physical $\hbar$. Fixing a nonzero charge product supplies k/c whether or not a theory quantizes charge. Charge quantization is an additional physical statement and does not follow from this dimensional argument. The normalized radiation unit is constructed separately in unit and indeterminacy.

A supplied least length with gravity

The declared constants (G, c, ℓ) admit the action monomial $c^3\ell^2/G$. A least-length premise therefore supplies an action unit; it does not prove a positive infimum of any specified action observable. Setting it equal to physical $\hbar$ is equivalent to supplying $\ell = \sqrt{\hbar G/c^3}$. Newton's optical fit interval, approximately 1/89000 inch, would instead give order 10^{22} J s in this formula, far from the quantum value. This dimensional comparison neither derives the least length nor makes the historical interval a universal cutoff. The Book I scholium's divisibility claim and its limits are treated with the held sources.

Admitted similarities and mechanical dilation

Similarity theorem. At fixed constants, suppose the admitted class contains $T_a s$ for arbitrarily small scale factors giving $A(T_a s) = a^\delta A(s) \to 0$, with $0 < A(s) < \infty$ and $\delta \neq 0$. Then its positive-action infimum is zero. If every $a > 0$ is admitted, the positive values are also unbounded above. A finite common value invariant under any admitted nontrivial rescaling must be zero.

Proof. Apply the displayed scaling to that one finite positive action. □

The map must preserve the admitted class. Changing units in the dimensional theorem and changing a physical model by a similarity are different operations. The following exact mechanical map makes that distinction explicit. For $m > 0$, use $T(p) = |p|^2/(2m)$ or $T(p) = \sqrt{m^2c^4 + c^2|p|^2}$, $c > 0$, and $V \in C^2(D)$. Put

$$D_a = aD, \quad V_a(Q) = V(Q/a), \quad q_a(t) = aq(t/a), \quad p_a(t) = p(t/a), \qquad a > 0.$$

Then $\dot{q}_a = \nabla T(p_a)$ and $\dot{p}_a = -a^{-1}\nabla V(q_a/a) = -\nabla V_a(q_a)$. Solution intervals scale by a; completeness, boundedness of an orbit and smoothness at each positive a survive. Mass, speeds and energy values are unchanged, while force and acceleration scale by a^{-1}. This is a map between potentials, not generally a symmetry of a single fixed Hamiltonian. Its phase map $F_a(q, p) = (aq, p)$ is conformally symplectic:

$$F_a^* \sum_i dQ_i \wedge dP_i = a \sum_i dq_i \wedge dp_i.$$

For corresponding closed orbits and finite segments,

$$J_a = aJ, \qquad \int_0^{aT} (p_a \cdot \dot{q}_a - H_a)dt = a \int_0^T (p \cdot \dot{q} - H)dt.$$

Push an invariant preparation forward by F_a. The identity $\phi_a^t F_a = F_a \phi^{t/a}$ proves invariance, and for a selected plane $Z = (x, p_x)$ with finite covariance,

$$\Sigma_a = \operatorname{diag}(a, 1)\Sigma \operatorname{diag}(a, 1), \qquad 2\sqrt{\det \Sigma_a} = a\, 2\sqrt{\det \Sigma}.$$

Uniform circular phase, subsystem independence and qualitative correlation decay survive. For stationary position increments,

$$\mathsf{h}_a(\Delta) = a\mathsf{h}(\Delta/a).$$

Corresponding windows $\Delta_a = a\Delta$ therefore scale by a; a finite long-window limit does too when the window limit is taken at fixed a first. No uniform fixed-window limit in a is asserted. A prescribed dimensional correlation time need not survive.

The admissibility tests are concrete: $V = -k/r$ maps $k \mapsto ak$; $V = -k/\sqrt{r^2 + b^2}$ maps $(k, b) \mapsto (ak, ab)$; a harmonic stiffness maps $k_s \mapsto k_s/a^2$. A fixed force ceiling excludes contraction of a nontrivial orbit. Fixed energy and speed ceilings do not. Excluding this map alone does not prove a floor, as the fixed-potential family below shows. Source/proof context: B30.

A prepared orbit: covariance and transport response

For any circular planar orbit of radius $R > 0$, momentum magnitude $P > 0$ and frequency $\omega > 0$, prepare uniform phase Φ:

$$q = R(\cos\theta, \sin\theta), \quad p = P(-\sin\theta, \cos\theta), \quad \theta = \omega t + \Phi.$$

Relativistic velocity gives $P = \gamma m R\omega$, $\gamma = (1 - R^2\omega^2/c^2)^{-1/2}$; Newtonian motion has $\gamma = 1$. The orbital action is $\ell = RP$. With $z = \omega\Delta$,

$$\mathbb{E}[x(t)x(t+\Delta)] = \frac{R^2}{2}\cos z, \qquad \mathsf{h}_x(\Delta) = \frac{\ell}{\gamma}\frac{1-\cos z}{z}.$$

Given $x \in (-R, R)$ the next positions are $x\cos z \pm \sqrt{R^2 - x^2}\sin z$, equally weighted, so

$$\mathrm{Var}(x(t+\Delta) \mid x(t) = x) = (R^2 - x^2)\sin^2 z, \qquad \overline{\mathsf{h}}_{\mathrm{cond}} = \frac{\ell \sin^2 z}{2\gamma z}.$$

Conditioning on the full canonical state gives a deterministic future. Both window coefficients vanish at zero and infinite window. Since $1 - \cos z \le \min(z^2/2, 2)$, a fixed fraction $\mathsf{h}_x \ge \eta\ell$, $\eta > 0$, requires $2\gamma\eta \le z \le 2/(\gamma\eta)$: its endpoint ratio is at most $1/(\gamma^2\eta^2)$. A broad plateau requires additional time scales.

For $Z = (x, p_x)$, uniform phase gives $\mathrm{Var}\, x = R^2/2$, $\mathrm{Var}\, p_x = P^2/2$ and zero cross covariance. Hence $2\sqrt{\det \Sigma} = RP = \ell$. Affine canonical changes within this plane preserve it, because $\Sigma \mapsto S\Sigma S^T$ and $\det S = 1$. On this orbit the flow is

$$S_t = \begin{pmatrix} \cos\omega t & (R/P)\sin\omega t \\ -(P/R)\sin\omega t & \cos\omega t \end{pmatrix}.$$

It preserves the determinant for every phase law, but a point-phase law has zero determinant despite $\ell > 0$. Equality with orbit action depends on preparation; nonlinear canonical invariance is not claimed. This is the rms-emittance construction with the stated normalization. Source/proof context: B29.

One smooth force-bounded Hamiltonian has zero orbital infimum

Fix $m, c, K, b > 0$ and

$$H = \sqrt{m^2c^4 + c^2|p|^2} + V(|q|), \qquad V(r) = Kb^2(\sqrt{1 + r^2/b^2} - 1).$$

This Cartesian-smooth confining potential has inward force $f(r) = Kr/\sqrt{1 + r^2/b^2} < Kb$. Finite energy bounds position and momentum, so trajectories are complete. Choosing $Kb \le F_{\max}$ meets a fixed force ceiling without changing the Hamiltonian across the family. No propagating mediator is constructed by this external-potential model.

At each $R > 0$, set $s = Rf(R)$ and

$$\gamma_R = \frac{s/(mc^2) + \sqrt{[s/(mc^2)]^2 + 4}}{2}, \quad v_R = c\sqrt{1 - \gamma_R^{-2}}, \quad P_R = m\gamma_R v_R, \quad \ell_R = RP_R.$$

The identity $mc^2(\gamma_R - \gamma_R^{-1}) = s$ is exactly the circular balance $P_R v_R/R = f(R)$. At fixed ℓ_R, the effective radial energy $U_\ell(r) = \sqrt{m^2c^4 + c^2\ell^2/r^2} + V(r)$ satisfies

$$U_\ell''(R) = f'(R) + (3 - v_R^2/c^2)f(R)/R > 0, \qquad f'(R) = K(1 + R^2/b^2)^{-3/2}.$$

Indeed the kinetic contribution is $P_R^2/(m\gamma_R^3 R^2) + 2P_R v_R/R^2$. Thus each circle is radially stable at fixed angular momentum, without Hamiltonian attraction. At $R \downarrow 0$,

$$s = KR^2 + O(R^4), \quad v_R \sim \sqrt{K/m}\, R, \quad \ell_R \sim \sqrt{mK}\, R^2, \quad \omega_R \to \sqrt{K/m}.$$

Both kinetic energy above rest and $V(R)$ are asymptotic to $KR^2/2$; force and acceleration tend to zero. Every strictly positive speed or acceleration upper ceiling holds eventually, and the period stays finite. Every member is regular, nonstationary and has positive uniform-phase covariance action, while the infimum is zero at this fixed Hamiltonian. Source/proof context: B31.

A sharp excitation-and-force bound for all closed regular orbits

Let $q : \mathbb{R}/T\mathbb{Z} \to \mathbb{R}^d$, $d \geq 2$, be C^2, with matching endpoint tangents and $v = |\dot{q}| \geq v_* > 0$. Set $p = P(v)n$, $n = \dot{q}/v$, where $P(v) = mv$ or $m\gamma v$; in the latter case assume $v < c$ and $v_* < c$. Suppose $|\dot{p}| \leq F_{\max}$ in this inertial frame. For scalar-potential mechanics this mechanical momentum is canonical. Vector-potential couplings require separate treatment.

Fenchel's total-turning theorem gives $\mathcal{K} = \int_0^T |\dot{n}|dt \geq 2\pi$, including self-intersecting regular closed curves. Orthogonality gives

$$|\dot{p}|^2 = \dot{P}^2 + P^2|\dot{n}|^2, \qquad |\dot{n}| \leq F_{\max}/P_*, \quad P_* = P(v_*).$$

Integrating and using monotonicity of $P(v)v$ proves

$$T \geq \frac{P_*\mathcal{K}}{F_{\max}}, \qquad J = \frac{1}{2\pi}\int_0^T P(v)v\,dt \geq \frac{P_*^2 v_*}{F_{\max}}\frac{\mathcal{K}}{2\pi} \geq \frac{P_*^2 v_*}{F_{\max}}.$$

The Newtonian value is $m^2v_*^3/F_{\max}$; the relativistic value is $m^2v_*^3/[F_{\max}(1-v_*^2/c^2)]$. A circle of radius $R_* = P_*v_*/F_{\max}$ at speed v_* attains equality. It is realized by a globally smooth force-bounded confining potential: choose a smooth nondecreasing χ equal to zero below $1/4$ and one above $1/2$, and set

$$V(r) = F_{\max}\int_0^r \chi(s/R_*)ds.$$

This is constant near the origin and linear at large radius, and has the required force at R_*. Sharpness is over the admitted potential class, not every preassigned potential. The positive speed floor is essential; letting $v_* \downarrow 0$ closes the bound. Oscillations with turning points fall outside its hypotheses. The distinct finite-duration reversal problem, with a sharp $1/24$ mesh defect, remains in bounded acceleration. Source reading: B32, Milnor Theorem 3.4 and its smooth-curve total-curvature identification (passages).

Singular and softened relativistic Coulomb cores

Fix $m, c, k > 0$, $r > 0$, and planar canonical Hamiltonian

$$H = \sqrt{m^2c^4 + c^2(p_r^2 + L^2/r^2)} + V(r).$$

A regular bound trajectory is complete and stays in a compact radial subset of $(0, \infty)$. Its angular cycle action is $\ell = |L|$; the spatial rosette need not close. Recoil, radiation and a dynamical second body are excluded. For $V = -k/r$, write $x = 1/r$ and

$$F_\ell(x) = \sqrt{m^2c^4 + c^2\ell^2x^2} - kx.$$

For $\ell > 0$, F_ℓ is strictly convex, starts with derivative $-k$ and has derivative limit $c\ell - k$. A finite minimum therefore exists exactly for $\ell > k/c$, at

$$E_{\min} = mc^2\sqrt{1 - k^2/(c^2\ell^2)}, \qquad r_0 = \frac{\ell\sqrt{c^2\ell^2 - k^2}}{mkc}.$$

The full bound domain is $E_{\min} \leq E < mc^2$. Equality gives the circle; otherwise the two radial turning points bound a compact allowed component. Energy bounds momentum there and the smooth vector field extends for all time. At $E \geq mc^2$ the outer radius is unbounded; below the minimum no state exists. At $\ell = k/c$, F_ℓ decreases to zero without attaining it; for $\ell < k/c$ it decreases to minus infinity, including $\ell = 0$. For the allowed plunging trajectories,

$$c^2p_r^2 = E^2 - m^2c^4 + 2Ek/r + (k^2 - c^2\ell^2)/r^2, \qquad \dot{r} = c^2p_r/(E + k/r).$$

If $\ell < k/c$, inward $|\dot{r}| \to c\sqrt{k^2 - c^2\ell^2}/k > 0$. At equality, allowed $E > 0$ gives $|\dot{r}| \sim c\sqrt{2E/k}\sqrt{r}$. Both reach the excluded centre in finite time. Thus the angular-action infimum is exactly k/c, excluded, with every larger value admitted continuously. It closes as $c \to \infty$ or $k \downarrow 0$; at $k = 0$ there are no nonstationary bound trajectories. At fixed ℓ, k, m, the Newtonian limits are $r_0 \to \ell^2/(mk)$ and $E_{\min} - mc^2 \to -mk^2/(2\ell^2)$.

Now fix a softening length $a > 0$ and $V_a = -k/\sqrt{r^2 + a^2}$. For every $\ell > 0$ the radial energy

$$U_{a,\ell}(r) = \sqrt{m^2c^4 + c^2\ell^2/r^2} - k/\sqrt{r^2 + a^2}$$

tends to infinity at zero and has expansion $mc^2 - k/r + \ell^2/(2mr^2) + O(r^{-3})$ at infinity. It therefore attains a minimum below mc^2 at finite positive radius. This gives a complete circle; energies between the minimum and mc^2 have compact allowed radial components. The positive angular-action infimum is zero at every fixed $a, m, k, c > 0$.

For fixed $0 < \ell < k/c$, minimizing radii tend to zero as $a \downarrow 0$. Choose a fixed large y with $c\ell/y < k/\sqrt{1 + y^2}$. Then $aU_{a,\ell}(ay)$ tends to a negative constant, so the minimum tends to minus infinity. On $r \geq \delta > 0$ the energy is bounded below by $mc^2 - k/\delta$, forcing its minimizer below every δ. The softened circles leave the singular model's regular domain: the two infima do not commute with removal of the core.

For elementary charges the singular threshold is $Z\alpha\hbar$. The Sommerfeld–Dirac circular-energy correspondence substitutes $\ell = \hbar|\kappa_D|$ into the displayed $E_{\min}$; radial quantum indicial coefficients similarly contain $c^2\ell^2 - k^2$. This correspondence uses a supplied angular quantum unit. Essential self-adjointness, distinguished extensions and endpoint dynamics have different thresholds; the precise Dirac and Klein–Gordon domains are in self-sufficiency, not inferred from the classical inequality. Sources: Boyer's equations (19)–(20), (37)–(42), read in B27; the internal Q14 comparison records the historical energy correspondence.

Consequence for STATE

Dimensional homogeneity limits the form of a possible universal floor; admitted similarities and the fixed smooth small-circle family defeat positivity under upper ceilings alone. The excitation bound and singular Coulomb threshold retain the exact extra premises that make them positive. A radiation preparation, pumped attractor, topological sector or record-risk constraint must be tested in its own admitted class. None of these classical results establishes quantum kinematics, universality of its scale, or the necessity of physical $\hbar > 0$.